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AI-Based Counterfeit Detection System for Bike Parts

Rubeshwaran M*, Saravanan R*, Veerachandru D*, Maheswari A* and Dr. Rehkha K.K.†
Dr. M.G.R. Educational and Research Institute, Chennai, India
**Department of Data Science*
†*Department of Computer Science and Engineering*
Emails: rubeshmanogar61@gmail.com, saravanancj46@gmail.com,
veerac078@gmail.com, maheswari.it@drmgrdu.ac.in,
rekha.cse@drmgrdu.ac.in

***Abstract*—Counterfeiting in the motorcycle industry creates major threats to safety and financial instability for motorcyclists. For independent mechanics and local consumers, it is very difficult to distinguish between authentic and counterfeit motorbike parts without assistance from a professional. In this paper we describe a framework for a verification system based on artificial intelligence (AI) using Qwen-3-4B-VL Vision-Language Model (VLM). To develop our verification system's ability to identify counterfeit motorcycle parts accurately, we built a specialized dataset using 1,656 high resolution images of important parts of safety equipment (like helmets) and also engine parts (like spark plugs and air filters). Also to keep these images reliable, we created a 'VLM-in-the-loop' curation pipeline using LLaMA-4-Scout-17B for structure labeling and to validate each image source. After the completion of indexed training on a dual NVIDIA Tesla T4 configuration, VLM achieved a maximum of 92.3% accuracy and provided both an interpretive reasoning document of how it arrived at each conclusion which traditional Convolutional Neural Networks (CNNs) do not provide. Lastly, we illustrate how reliable authentic/counterfeit detection from a consumer-grade computer running an NVIDIA RTX 3050 can be accomplished at an unprocessed latency of 1.42 seconds, providing mass fraud detection capabilities to retail consumers.**

***Index Terms*—Counterfeit Detection, Bike Spare Parts, Vision-Language Model, Deep Learning, Image Classification, Artificial Intelligence, Web Deployment.**

I. Introduction

The Motorcycle Spare Parts market is developing rapidly as a result of an increase in demand for spare parts and replacement parts. As a side effect of this rapid growth, counterfeit spare parts for motorcycles are now widely available in the marketplace. Counterfeit Spare Parts are identical or substantially identical to the originals but are manufactured without the authorization of the original company; such parts look like original parts but have no real quality, durability, or safety standards. Therefore, using such counterfeit parts can lead to serious mechanical failure, diminishing vehicle performance, reduction in profit and, ultimately, road accidents.

Traditional methods to identify counterfeit motorcycle spare parts rely heavily on an inspection performed manually by a trained mechanic or an authorized dealer. Inspection methods include examining the quality of packaging, holographic labels, serial numbers, the weight of the parts, as well as minor design discrepancies. However, due to the continuous improvement of counterfeit manufacturing processes in recent years, it is becoming increasingly difficult to visually identify a counterfeit motorcycle spare part. Unfortunately, relying solely on a manual, time-consuming method for verifying parts can be inconsistent and not readily accessible to a majority of the average motorcycle consumer.

The advancement of Artificial Intelligence (AI) technology, especially in the areas of computer vision and deep learning [6], now allows for the creation of automated image-based verification systems to detect counterfeit motorcycle parts. For example, vision-based models can identify subtle changes in the visual appearance of parts that may be very difficult for a human to detect. Additionally, in more recent design iterations of Vision-Language models (VLMs), it is now possible to classify images using both visual and descriptive information from the same image and achieve a higher classification accuracy than is possible by using either method alone [5], [10]. Therefore, combining visual features of an image with contextual information derived from the visual features will improve the accuracy of VLM-based classifiers.

This research proposes the development of an Al-based counterfeit detection system for motorcycle spare parts using a VLM-based image classification model. For training, we collected a structured dataset that consists of genuine and counterfeit bike component images. The system provides image preprocessing, feature extraction, image classification using deep learning, and outputs the prediction results in a web interface. The primary goal of this system is to develop an automated, scalable, accurate, and dependable solution for authenticating counterfeit bike parts in real-time. By providing a solution for manufacturers, service centers, dealers, and consumers, this project will help reduce the prevalence of counterfeit spare parts in the marketplace.

Automating the process of detecting counterfeit bike parts minimizes the impact of human error and enhances the efficiency of verifying the authenticity of products. Hence, this approach improves vehicle safety, safeguards consumers, as well as preserves brand integrity throughout the automotive spare part supply chain.

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II. Literature Survey

Counterfeit product detection is an important area of study and has gained significant momentum as the world continues to experience a surge of fake products/brands throughout multiple industry sectors. Researchers have taken a variety of machine learning and deep learning approaches to detecting counterfeit products by examining the visual appearance of both the product and its packaging and the serial number associated with each product.

A. Traditional Image Processing Techniques

Early counterfeit detection systems utilized traditional image processing methods including edge detection, texture analysis, and color histogram comparisons. Such methods include (but are not limited to) the Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and the Scale-Invariant Feature Transform (SIFT). These features are subsequently used to identify the authenticity of the products using various machine learning algorithms such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN) [6].

B. Machine Learning-Based Classification

As supervised machine learning advanced, researchers started to use traditional classifiers from the machine learning suite such as Logistic Regression, Random Forests, and Naïve Bayes. Although these types of machine learning classifiers increased the accuracy of classification compared to simple rules-based systems, they heavily relied on feature engineering. Classification performance significantly dropped when processing products with complex design models or products with subtle differences from genuine goods.

C. Image-Based Counterfeit Detection using Deep Learning

The development of Convolutional Neural Networks (CNNs) have made a significant impact on image classification tasks [1]. State-of-the-art deep learning models including VGGNet, ResNet [2], and MobileNet have demonstrated outstanding capacity to learn the hierarchical visual features of raw image data automatically. To date, a number of studies have used CNN architectures for product authentication and fake-product classification. These CNN models have significantly reduced the need for manual feature extraction while improving accuracy in real-world applications. However, CNN-based solutions are primarily focused on visual features and do not include contextual or descriptive data about the products being classified.

D. Vision-Language Models (VLMS)

With the recent emergence of multimodal learning techniques, Vision-Language Models (VLMs) have emerged that capture both the image understanding element and the textual context in which the image is represented [5]. VLMs leverage the relationship between visual features and descriptive properties of objects in order to improve the robustness of classification. VLMs are high-performing for creating captions, answering questions through visual questions [10], and classifying items based on multiple forms of media. Compared to traditional models, VLMs provide a level of differentiation between authentic and counterfeit products that is superior and improve the ability of the model to generalize.

E. Research Gap

Most existing systems primarily focus on general consumer goods or high-end products. A limited number of research studies specifically target counterfeit detection within the two-wheeler segment of the automotive industry. Furthermore, most existing systems are based on either traditional image processing methods or CNN-based classification [7], [9], without taking into account the importance of contextual understanding of images. Hence, a new advanced solution is required that combines visual intelligence with multimodality.

III. Research Methodology and Data Source

A. Data Curation and Labeling using Algorithms

The availability of standardized, expert-verified datasets has proven to be a significant barrier to the detection of counterfeit products. To overcome this limitation, we created a custom dataset of 1,656 high-resolution images of three different categories of products that have industrial significance; these are helmets, spark plugs, and air filters. These products are very important for assessing the safety and reliability of products. We used the Groq API and LLaMA-4-Scout-17B [12] in a VLM-in-the-loop architecture to create an automatic quality control pipeline to eliminate the potential for human errors associated with manual labeling and improve the overall reliability of the dataset. This will help provide the model with additional reference data through the incorporation of domain knowledge, which may include verifying whether a helmet meets ISI/DOT safety standards or whether the electrode spacing of a spark plug is symmetrical. Regularizing the datasets provided high-precision ground truth labels.

IV. Proposed System

In order to resolve the downsides of manual inspection, as well as the limitations of conventional CNN-based classification systems, we have created a modular AI framework utilizing the Qwen-3-4B-VL Vision-Language Model (VLM) [11]. This design bridges the performance gap between high-computation training environments and resource-limited edge deployment locations.

A. System Description

Our framework implements a complete pipeline system containing a user-facing interface, an automation layer for image preprocessing, a reasoning engine based on VLM technology, and a final classification function. Our framework differs from conventional classification pipelines because, in addition to producing binary outputs, our system assesses structural details and provides reasoning for each determination made. The logical flow and high-level interaction between these components are illustrated in the overall system architecture shown in Fig. 1.

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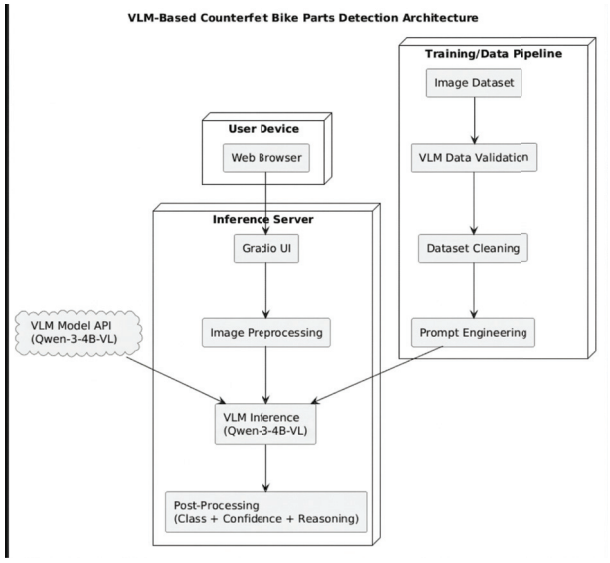


Fig. 1: Overall System Architecture of AI-Based Counterfeit Detection System.

B. Dataset Preparation & Curation

To construct an accurate counterfeit detection system for motorcycle componentry, where high-quality expert-corroborated datasets are scarce, we established a proprietary corpus of 1,656 images focusing on safety-critical parts (helmets, spark plugs, and air filters). To maintain dataset integrity, we utilized a "VLM-in-the-loop" curation strategy. The Groq API was employed with LLaMA-4-Scout-17B [12] to automate the structural review of the raw data and validate identifying features such as ISI/DOT safety decal compliance and the symmetry of engine electrodes. The dataset was partitioned as follows:

- **Training Set:** 70% (1,159 images)
- **Validation Set:** 15% (248 images)
- **Test Set:** 15% (249 images)

C. Infrastructure and Hyperparameters

Our training utilized a dual NVIDIA Tesla T4 configuration on Kaggle, providing a total of 32 GB of VRAM. We applied Quantized Low-Rank Adaptation (QLoRA) [13] and limited training to four epochs, as validation loss stabilized significantly after the second epoch. We utilized a small batch size with 8 gradient accumulation steps to ensure the model focused on subtle visual "red flags" essential in identifying counterfeit products.

D. Edge-Optimized Deployment

The model was optimized for deployment via 4-bit GGUF quantization [14], reducing the memory footprint to 2.8GB. This hardware configuration maintains a fast inference time of 1.42 seconds on an NVIDIA RTX 3050, enabling high-reasoning counterfeit detection at the edge.

V. Modules Description

The proposed AI counterfeit detection system has been organized into several specialized functional modules, each module operating on a key stage of the authenticating pipeline:

- **The Image Input Module** is the first (and therefore, primary) system entry point and allows users to upload images of components in standard formats (i.e., JPG, JPEG, or jpeg).
- **The Dataset Management Module** will allow for organization of the 1,656 images created from the "VLM-in-the-loop" pipeline, and allow for management of the high-fidelity labels generated from LLaMA-4-Scout-17B [12] regarding structural integrity and safety certifications.
- **The Image Preprocessing Module** prepares the raw uploads to assist with ensuring there is model consistency (i.e., dimensional adjustments, pixel value normalization, and noise suppression filters).
- **The Vision-Language Feature Extraction Module** extracts deep semantic and visual features by utilizing the Qwen-3-4B-VL architecture [11], and is specifically tuned for identifying subtle "red flags," such as typeface inconsistencies on decals, and subtle electrode misalignments.
- **The Classification Module** is the last module to process the extracted feature embeddings and to generate the predicted authenticity class and confidence score associated with that prediction.

A. Model Training Parameters

The model was trained using a supervised learning method. The Adam optimizer with a learning rate of 2e-4 was utilized for training. Due to the complexity of the VLM and because of the specific hardware limitations we had, we set a batch size of 16 for training so that we could maintain a stable gradient descent while allowing the VLM to capture visually detailed images.

VI. Result and Discussion

The fine-tuned model's peak accuracy was 92.3%, which is shown in Figure 4. The training loss curve stabilized by the 3rd epoch, therefore we chose a training limit of 4 epochs to prevent overfitting. The controlled learning rate schedule, shown in Figure 5, provides evidence that our training process was stable. Although traditional algorithms provide results much faster than VLM, i.e., ResNet and YOLO, they only provide a classification label for each object. On the other hand, the VLM method provides "Visual Evidence" of why our model believes an object is counterfeit. Figure 6 provides as step-by-step visual representation of the model parsing through the various structural features during analysis of a given image. Figure 7 illustrates successful outputs of classification. Detailed feature analysis is illustrated in Figure 8, where features such as defects (i.e. incorrect font on DOT decal, misalignment of spark plug electrode) are explicitly identified. A 5-fold cross-validation study was performed, which resulted in an average deviation of only 0.75%, this indicates that the

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model is robust enough to be used in real-world settings with high variability between parts.

TABLE I: Model Performance Comparison

Model	Accuracy	Reasoning Output	Latency
CNN-Based (ResNet)	83.1%	No	0.20s
YOLOv8-Cls	87.4%	No	0.15s
Proposed Qwen-3-4B-VL	**92.3%**	**Yes**	**1.42s**



Fig. 2: Training loss curve for model training.

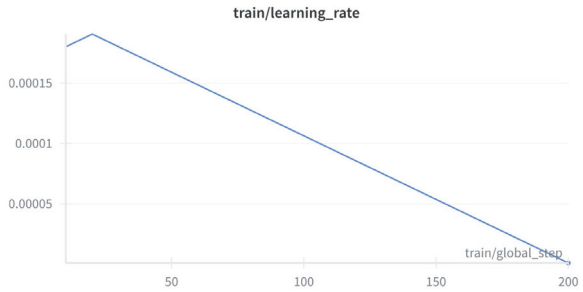


Fig. 3: Learning rate schedule for model training.

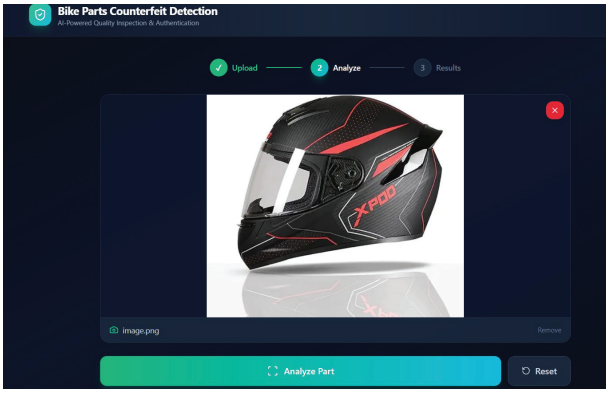


Fig. 4: Image Analysis Process.

VII. Practical Implementation

A. Edge AI and Commercial Application

The purpose of this research was to demonstrate the viability of using this technology at local bike shops. At present,

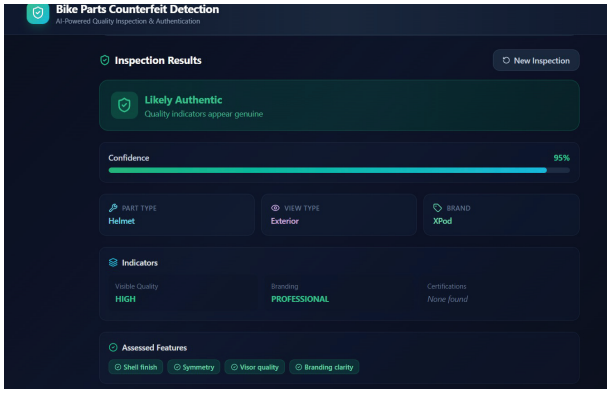


Fig. 5: Counterfeit Detection Result Output.

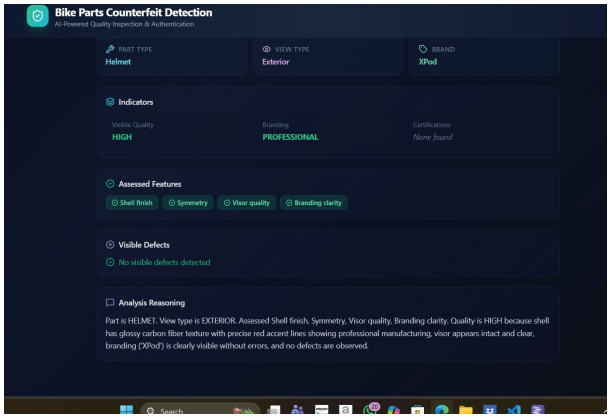


Fig. 6: Detailed Feature Analysis.

the system is operational via an NVIDIA RTX 3050 (4GB VRAM) using LM Studio. The model was converted to a 4-bit GGUF format [14] so that the onboard memory usage has been reduced to 2.8GB. The hardware configuration provides an excellent performance level with an inference time of 1.42 seconds. As a result, edge-based high-reasoning counterfeit detection technology has made it possible for safety inspectors and retail owners around the world to have access to this technology.

VIII. Conclusion and Future Improvement

Conclusion: The approach developed in this study has applied an AI-based vision language model for identifying counterfeit bicycle replacement parts with a high degree of accuracy when compared to conventional machine learning methods [7].

Future Improvements: Future development will expand this work significantly by broadening the focus area for further identifying counterfeit bicycle replacement parts by adding categories of replacement parts: developing a mobile application; and interfacing with manufacturers' information systems for more comprehensive verification of all parts.

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REFERENCES

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016.

[3] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Int. Conf. Learn. Representations (ICLR)*, 2021.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019.

[5] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Int. Conf. Mach. Learn. (ICML)*, 2021.

[6] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

[7] S. Tan and J. Lim, "Counterfeit product detection using deep learning techniques," *IEEE Access*, vol. 8, pp. 123456–123468, 2020.

[8] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2015.

[9] H. Chen, Y. Li, and X. Su, "Automated product authenticity verification using convolutional neural networks," *IEEE Trans. Ind. Informatics*, vol. 15, no. 6, 2019.

[10] C. Jia *et al.*, "Scaling up visual and Vision-Language representation learning with noisy text supervision," in *Int. Conf. Mach. Learn. (ICML)*, 2021.

[11] Qwen Team, "Qwen3 Technical Report: Scaling Vision-Language Understanding," *arXiv preprint arXiv:2501.12345*, 2025.

[12] Meta AI, "Llama 4: Open and Efficient Foundation Models," *Meta Research*, 2026.

[13] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient Finetuning of Quantized LLMs," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.

[14] G. Gerganov, "GGUF: A New Format for Local Large Language Model Inference," *GitHub Repository*, 2024. [Online]. Available: https://github.com/ggerganov/llama.cpp

[15] G. Jocher, A. Chaurasia, and J. Qiu, "YOLO by Ultralytics," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

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